

Instructions:

You've been assigned to remotely investigate a compromised Windows workstation as part of a live incident response scenario. The affected system has been imaged and all necessary forensic artifacts have been gathered. Your objective now is to investigate the scope of the compromise and identify any persistence mechanisms installed by the attacker.

Use your endpoint analysis skills to conduct the investigation and answer the questions below.

Challenge File:

03_Endpoint_Security/Windows/Challenges/challenge.exe

To start the challenge, execute challenge.exe in an Administrator-level command prompt. Make sure to not close the program or window until after you've completed the challenge. However, you can temporarily minimize it. After running the executable, you can open additional command prompt windows to complete the investigation.

After completing the challenge, you can clean up and restore the system by terminating the challenge.exe process (Press Ctrl + C). Then, run challenge.exe -revert.

Q1 Simulate the compromise by running challenge.exe in an Administrator-level command prompt. Make sure to not close the program or window until after you've completed the challenge. However, you can temporarily minimize it. Enter "done" to continue.

done

Q2 What port is the challenge.exe malware listening on?

50050

Q3 What is the Process ID (PID) of the malware? This answer cannot be verified, but it's required for the next question(s).

3800

Q4 List all of the loaded DLL modules for the listener's associated process. What are the names of the two DLLs that start with the letter m?

msvcrt.dll, mswsock.dll

Q5 What is the name of the malware's parent process?

cmd.exe

Q6 List all the shared resources on the local system. What is the name of the share that the attacker created?

xkalibur

Q7 What path or resource does the attacker's share point to?

C:\Users\tcm\AppData\Local\Temp\46d5b8556d0d3e30ec1

Q8 The attacker created a Run entry to establish persistence on the system. What is the full path of the registry key?

HKCU\Software\Microsoft\Windows\CurrentVersion\Run

Q9 What is the name of the malicious entry the attacker created in the previously identified registry key?

CleanUpController

Q10 What is the full file or image path that this registry entry points to?

C:\Users\tcm\Downloads\wininit.exe

Q11 What is the name of the backdoor service installed by the attacker?

WindowsActiveService

Q12 What is the START_TYPE configuration of the service?

autostart

Q13 What is the full path to the service binary?

C:\Users\tcm\Documents\svcbackdoor.exe

Q14 What is the name of the scheduled task created by the attacker?

\ayttnzc

Q15 What is the full path to the executable that the scheduled task runs?

C:\Users\tcm\Downloads\beac0n.exe

Q16 At what time is the scheduled task set to run?

3:30:00 AM

Q17 Excellent work! Clean up and restore the system by terminating the challenge.exe process (Press Ctrl + C). Then, run challenge.exe -revert. Afterwards, enter "done" to continue.

done

Challenge URL

<https://challenges.malwarecube.com/#/c/592319ba-7fd3-4322-ad14-1fa946fedc46>